

A TCN-based Vortex-Induced Vibration Detection by GNSS-IMU Fusion for Sea-Crossing Bridge

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Abstract—Monitoring the structural health of bridges, especially sea-crossing bridges, is vital for their longevity and safety. Among the potential risks during the operation of sea-crossing bridges, vortex-induced vibration (VIV) has drawn significant attention due to its potential to cause collapse. To accurately detect VIV, introducing new sensors such as GNSS has become a trend. However, current methods often suffer from high latency and low reliability. Additionally, the integration of GNSS poses significant challenges to power supply. To address these challenges, this paper proposes a novel deep learning-based adaptive GNSS-IMU fusion method for low-latency, high-accuracy, and energy-efficient detection of bridge VIV. We have employed a lightweight Temporal Convolutional Network (TCN) to replace traditional VIV detection methods, significantly reducing the latency of VIV detection. Additionally, Our method is the first to leverage the unique reflection mechanisms of the ocean surface to model GNSS multipath effects, significantly enhancing the accuracy of adaptive data fusion. Finally, the adaptive fusion mechanism has also substantially decreased the power consumption associated with integrating GNSS. The method is validated using both synthetic data and real vibration testbed data. Experimental results demonstrate that, compared to traditional VIV monitoring methods, our approach reduces detection latency from minutes to seconds while maintaining high detection accuracy. Under the same power supply conditions, the deployment duration is extended by 41.8%. We have also designed an IoT-based bridge Structural Health Monitoring (SHM) station to execute our algorithm.

Index Terms—Adaptive Fusion, Multipath, Deep learning, Vortex-induced vibration, Sea-crossing Bridge

I. INTRODUCTION

The increasing adoption of IoT (Internet of Things) devices in infrastructure has significantly enhanced the capabilities of structural health monitoring (SHM) in recent years [1].

Bridges, as one of the most critical transportation infrastructures worldwide, have largely adopted IoT-based SHM networks to ensure their safety and normal operation. The primary task of these monitoring networks is to detect bridge displacement and deformation, including the bridge box girders [2] and suspension cables [3]. Among these, wind-induced vibration is

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a common phenomenon, with vortex-induced vibration (VIV), which can be formulated as a harmonic motion [4], being the most severe. Extreme cases of VIV can lead to catastrophic outcomes. In 2020, the Humen Bridge in China experienced VIV for over 20 hours, resulting in displacements exceeding 30 cm and forcing a 10-day closure. These incidents underscore the critical importance of precise VIV monitoring.

Traditional methods for detecting VIV in bridges fall into two categories. The first approach is amplitude detection using displacement sensors [5], where VIV is identified by comparing measured amplitudes against predefined thresholds determined during the bridge design phase. Recent advancements in Global Navigation Satellite System (GNSS) technology, particularly GNSS-RTK (Real-Time Kinematic) systems [6], have enabled centimeter-level accuracy, making GNSS a complementary tool for amplitude monitoring in SHM. However, this method faces limitations in threshold accuracy and sensor deployment for large-span cross-sea bridges, where extensive sensor coverage is costly and laboratory-derived thresholds often deviate from real-world conditions. The second method, based on accelerometers or IMUs [3], [7]–[9], employs the Hilbert transform, as shown in Fig.1(a) to analyze acceleration data for VIV detection. Nevertheless, the edge effects [3], [7] of the Hilbert transform cause high detection latency and missed VIV events. These methods share a common limitation: due to the different deployment locations and altitude of the sensors, there is no hardware foundation for effective data fusion.

Inspired by the above discussion, simultaneously monitoring both the amplitude and pattern of vibrations can significantly improve the detection rate of VIV. Given that GNSS and IMU can be deployed at the same location, fusing GNSS-RTK and IMU data [10] presents a feasible and effective approach. However, this approach also faces significant challenges in sea-crossing bridges. First, adding GNSS data does not change the VIV detection method, leaving **high latency and missed VIV events unresolved**. Second, GNSS signals are prone to environmental interference, such as **multipath effects, making the GNSS data unreliable**, shown in Fig.1(b). Third, GNSS devices consume far more power than IMUs and displacement sensors, **failing to meet endurance requirements in emergencies**.

To address the aforementioned issues, we propose an adap-

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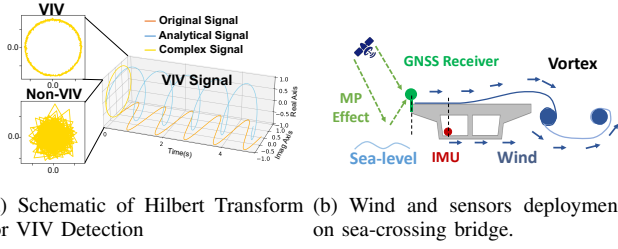


Fig. 1: 1(a) illustrates the Hilbert transform VIV detection. 1(b) presents a schematic of the vortex-induced vibrations that may occur as the wind sweeps across the bridge girder box.

tive fusion algorithm for VIV monitoring based on Temporal Convolutional Networks (TCNs), which fuses GNSS-RTK displacement data with IMU acceleration measurements. By employing the deep learning-based approach, our method significantly reduces VIV detection latency from minutes to seconds while maintaining high detection accuracy. Furthermore, to address the unique challenges of marine environments, we introduce a GNSS confidence evaluation mechanism that accounts for signal fluctuations caused by the multipath effect of sea surface reflections [11]. This mechanism ensures robust and accurate data fusion even in maritime conditions. Additionally, the adaptive fusion of IMU and GNSS data not only maintains high precision but also reduces power consumption, making the system more energy-efficient and sustainable for long-term deployment.

The key contributions of the paper are as follows:

- We propose an **adaptive fusion** based on the IMU and GNSS-RTK data, adding the GNSS multipath-effect signal quality trigger into it, achieving a balance between energy consumption and monitoring accuracy.
- We designed a **lightweight TCN model for detecting VIV patterns**, which can accurately detect VIV in sea-crossing bridges.
- We designed an **IoT-based bridge SHM system for the target bridge and deployed our algorithm on it**. This implementation serves as a reference for the future development of similar IoT-based SHM devices.

II. METHODOLOGY

A. System Overview

Fig.2 shows the birds-eye view of the Adaptive-Fusion bridge VIV monitoring system architecture. The system consists of three main components: **Data Acquisition Part**: This part takes the responsibility for data reception, pre-processing, and hardware control. **Adaptive-Fusion Part**: This part is the core part of our system. It merges the IMU acceleration data with displacement data adaptively and detects whether VIV occurred. This part also includes the GNSS multipath evaluation module, and the deep learning VIV pattern detection module. **Result Part**: This part precisely derives both the longitudinal displacement data and VIV detection results.

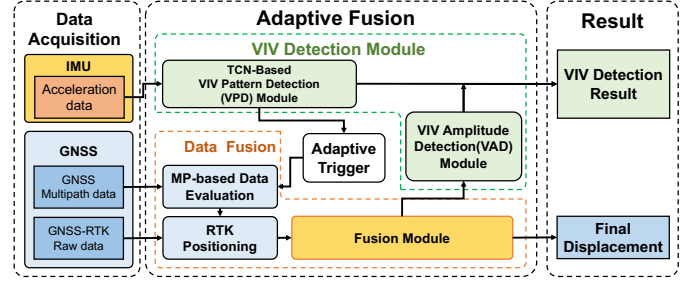


Fig. 2: System Overview. The Core part is the Adaptive trigger part. It gives the trigger to the active GNSS and Fusion.

B. Multipath-based GNSS Positioning Optimization Method

We first model the multipath effects of GNSS. GNSS errors are typically categorized into three main types: satellite errors, path errors, and receiver errors. Unlike many other error sources that can be mitigated through various techniques, multipath error multipath error arise when a periodic GNSS signal reaches the receiver via multiple reflective paths, leading to signal distortion. which is notoriously difficult to eliminate entirely.

To simplify the modeling process, we extracted the two most significant components of multipath error [12]–[14]: pseudorange error P , which is used for coarse positioning, and carrier-phase error C , which is essential for precise positioning.

Then pseudorange and carrier phase multipath errors existing in the observation date, represented as MP , can be described in the form of additive errors, which could be written as [14]:

$$MP = P + C \quad (1)$$

Based on the periodic pattern of multipath signals, we can extract the intensity of the multipath effects over a specific duration from the multipath combination. In this study, a three-minute window is utilized to ascertain the confidence level of the observed measure. The strength of the multipath for a set of signals can be obtained by determining its upper and lower envelope curves. These envelope curves are acquired by respectively identifying the maximum $Max(MP^t)$ and minimum $Min(MP^t)$ values of the multipath combination for each timestamp t .

$$H = \frac{\sum_{t=0}^n [Max(MP^t) - Min(MP^t)]}{n} \quad (2)$$

$$Q = \operatorname{argmax} \frac{1}{\sum_1^i (H_i)} \quad (3)$$

where H is the average envelope range of the multipath effect for each satellite. Q is the multipath effect confidence coefficient of all the satellites captured in the current GNSS signal. When the H becomes larger, the Q will be closed to 0, and vice versa. We only need to find the smallest $\sum_1^i (H_i)$ combination within the current observation time period, which means calculating the max value of Q . This allows us to

identify the best satellite combination, which can then be input into the RTK positioning module for calculation. Thus, we effectively utilize the multipath effect to filter satellites and enhance positioning accuracy.

C. Adaptive Fusion

Our primary objective is to devise a high-precision bridge VIV detection method. As previously discussed, VIV detection necessitates the identification of the vibration pattern and amplitude. To make it simple, this paper primarily focuses on longitudinal Z-axis data for VIV monitoring. While IMUs, with their high frequencies, can accurately monitor vibration patterns but tend to drift when integrating displacements. Conversely, GNSS-RTK can precisely monitor displacements. However, the GNSS receivers installed on sea-crossing bridges typically sample at a rate of 1Hz, which cannot capture the vibration pattern according to the sampling theorem. Therefore, the most commonly used method for merging the strengths of both sensors is the Extended Kalman Filter (EKF) [15].

A key metric within the EKF framework is the variance of the measurements and observations. This directly influences the fusion results. Relying only on the declared errors from the sensors as input is not representative of real bridge conditions. Therefore, using the GNSS data quality estimation method discussed in the previous section, we aim to achieve optimal results by adjusting the variance of the GNSS measurements.

D. Deep Learning based VIV Pattern Detection

Compared to the complex nonlinear fitting tasks in other machine learning tasks, harmonic motion of VIV is relatively simple. This simplicity allows the final training model to be **lightweight** while still capable of fulfilling our task requirements. Hence, we select the TCN (Temporal Convolutional Network) [16]. This model is originally designed for time-series data, and [16] demonstrates that TCN has better performance in time-series data processing tasks compared to traditional RNNs (Recurrent Neural Networks), CNNs (Convolutional Neural Networks), and more lightweight than Transformers. The original design of TCN addressed many complex time-series prediction problems. In this paper, TCN is slightly modified to become a classification model, used to identify VIV patterns and to check whether they match the VIV of sea-crossing bridges.

Given the requirement for the deep learning model to be lightweight and real-time, we streamlined the TCN model by reducing its residual modules by half. We found that for the task of VIV pattern detection, this modification may slow down the training speed but does not affect the final prediction outcome.

E. Adaptive VIV Detection

Based on the aforementioned insights, we implement the Adaptive-Fusion methodology. The VIV amplitude detection method is based on [3].

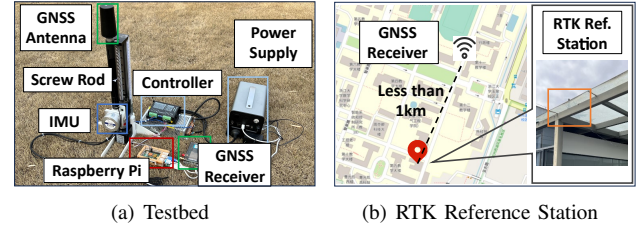


Fig. 3: Testbed and RTK Reference Station

III. IMPLEMENTATION

A. Testbed

This section delineates our experimental platform. As illustrated in Fig.3, we construct a vibration testbed capable of precise displacement control to simulation of motions on bridges, driving a 1650-threaded rod with a length of 200 mm.

At the core board of our testbed is the Raspberry Pi 4B. For inertial measurement, the Beiwei IMU200 is selected, while the Sino K823 has been employed both as our GNSS receiver and for the RTK reference station. In practical deployment, as shown in Fig.9, the core board will be replaced with the RK3568, and the GNSS receiver will be substituted with a product of equivalent quality. The reference station has been positioned atop our academic building, as shown in Fig.3(b). This testbed also serves as the sample for the monitoring station we will deploy in the target bridge.

B. Dataset

The datasets (except the multipath effect validation) utilized in this paper are presented in Table.I.

TABLE I: Datasets

No.	Data Type	Data Duration	Data Explanation
1	G	8 Hours	include 50 VIV Events
2	G	8 Hours	include 100 VIV Events
3	G	1 Hours	include 3 VIV and 1 Non-VIV
4	M	8 Hours	include 33 VIV Events

All datasets include IMU and GNSS data, along with Ground Truth data. *G* means generated data. *M* means measurement data.

Datasets 1 to 3 are generated and combined using the `gnss-ins-sim` tool, based on specific motion states of random vibration and VIV. These generated data consider the real bridge state, with an IMU sampling rate of 50 Hz and a GNSS sampling rate of 1 Hz. In Datasets 1 to 2 the VIV motion frequency is set to 0.8 Hz [17]. The VIV amplitude for Dataset 2 is set to 170 mm. Dataset 3 is primarily to validate the robustness of the adaptive algorithm. It contains three VIV with an amplitude of 200mm, two of which have a frequency of 0.8Hz, and one has a frequency of 0.2Hz. It also includes one non-VIV with an amplitude of 100mm and a frequency of 2Hz, which represents the short-duration, small-displacement resonances often caused by vehicular passage on bridges. Dataset 4 is similar to the Dataset 1 and 2, but

combined with the measurement VIV data from the testbed rather than the generated data.

C. Energy Consumption Metric

In subsequent experiments, for the purpose of simplifying the criteria for evaluating energy consumption, the following assumptions are made based on the 'Adaptive Fusion' section: 1) The operational power consumption of the IMU, denoted as P_I , is 1 watt; 2) The operational power consumption of the GNSS, denoted as P_G , is 3 watts; 3) The operational power consumption of the computation board, denoted as P_{core} , is 3 watts; 4) The total energy supply capacity, denoted as W_{system} , is $0.5 \text{ kW} \cdot \text{h}$, consistent with the power source utilized in our experimental testbed.

These assumptions are utilized to estimate the system's lifespan, t , when it is entirely disconnected from any power source.

D. TCN Model Training

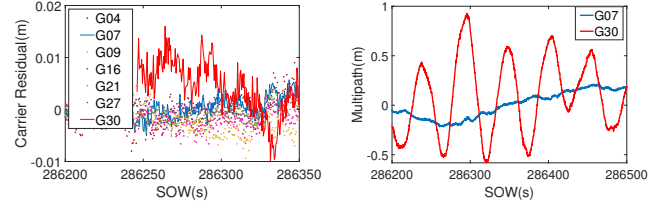
We primarily utilized the IMU data from a subset of Dataset 4 to train our streamlined TCN model in the VPD module. The training parameters involved segmenting the Dataset 4 data into window size of 600 data points (12 seconds), with window segmentation every 30 data points. This means each data point could be used by a window up to 20 times. Training is conducted on a desktop workstation equipped with an AMD R7 5800 CPU, 32GB of memory, and an Nvidia GTX 1080Ti graphics card.

IV. EVALUATION

A. Verification of Multipath effect in GNSS signal

The GNSS data used in this evaluation section is collected from the GPS (Global Positioning System, one of GNSS) during an approximately one-hour observation on a traffic-loaded bridge that began on SOW (Seconds of GPS Week, a kind of timing method of GNSS) 285200, GPS week 2247. The bridge is oriented in an east-west direction and the antenna is fixed to the northern guardrails. The specific placement of the GNSS antenna can be referred to in schematic Fig.1(b). The impact of multipath effects on the GNSS positioning accuracy could be reflected in the residual data obtained from RTK positioning results. This is because, under the premise of accurate positioning, the value of the carrier residuals represents the degree of interference existing in the observation data Fig.4(a) depicts the situation where G30 (The 30th satellite of GPS) passed the elevation mask test for the first time and is incorporated into the resolution process at about SOW 286300. Compared to other satellites with relatively small residual intervals, the residual distribution of G30 is considerably larger.

As we illustrated before, significant error components in the observations could have been caught in the early pre-process step. At epoch SOW 286200, G30's signal is subject to multipath interference caused by strong reflection from the water surface, resulting in regular periodic components outside



(a) Carrier-phase residuals in RTK, (b) Multipath linear combination of SOW 286200, GPS Week 2247 G07 and G30, SOW 286200, GPS Week 2247.

Fig. 4: Multipath effect for GNSS satellites. G30 have a significant multipath effect.

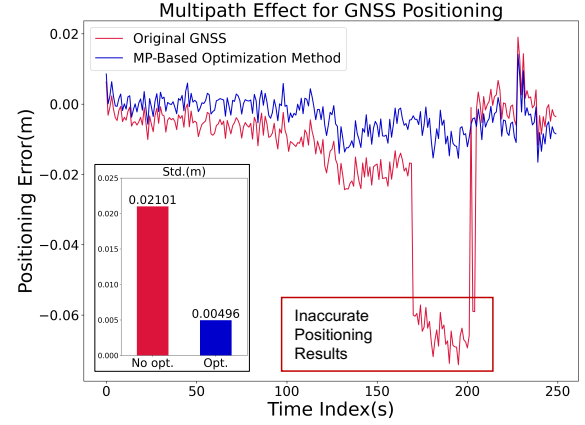


Fig. 5: Result of Multipath-based GNSS Positioning Optimization.

the direct signal, while the multipath of G07 is much smaller. This shows that our method is theoretically feasible.

Therefore, we applied the multipath-based (MP-based) optimization methods. In Fig.5, between time steps 150 and 200, due to severe multipath effects on several satellites (such as the G30 satellite in Fig.4(b)), the positioning results gradually deviated from the median and ultimately led to a failure in RTK positioning, significantly affecting measurement accuracy. However, the MP-based optimization methods demonstrated more robust positioning results and stronger resistance to multipath interference. In the lower-left corner of the figure, the standard deviations of the two data sets are supplemented, showing that the MP-based optimization method significantly improves positioning accuracy.

B. VIV Detection Rate

We compared a total of four experimental approaches: **IMU-Integration (IMU)** [3], [7]: It derives the displacement by integrating the raw IMU acceleration data twice. **GNSS-RTK-Only (GNSS-RTK)** [6]: This method directly utilizes the raw RTK results. **Kalman-Filter (KF)** [10]: This approach fuses data from both the IMU and GNSS-RTK. **Adaptive-Fusion (AF,Ours)**: The method proposed in this paper.

In the Evaluation section, figures and tables will reuse the experiment numbering and descriptions. The VIV detection

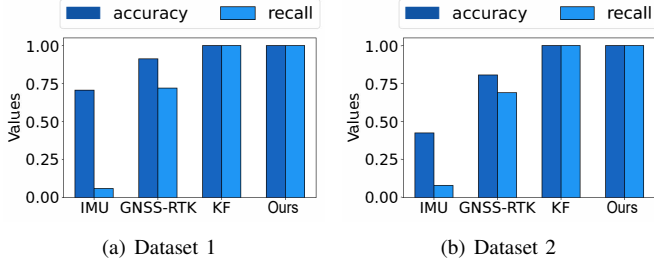
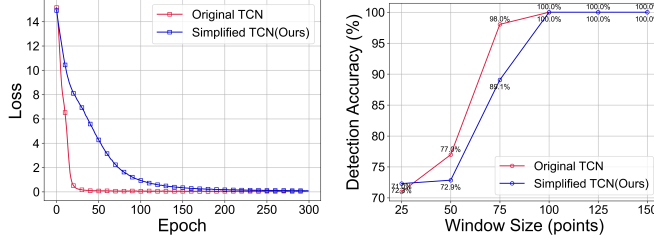


Fig. 6: Our method applied for Different Dataset.



(a) Training time comparison between original TCN and Simplified TCN (Ours) (b) VIV Detection accuracy comparison between original TCN and Simplified TCN (Ours)

Fig. 7: Evaluation for TCN.

rate serves as a direct metric to judge the accuracy of our methods.

As shown in Fig.6, IMU performs the worst, then GNSS-RTK performs better, and KF and AF (Ours) are similar.

The result is intuitive. IMU result, affected by truncation errors in integration, cannot consistently restore long-term displacements even with calibrated values. GNSS-RTK, with its low sampling rate, cannot fully capture the modal shapes. Both KF and AF (Ours) can precisely detect VIV.

As demonstrated in the shown in Fig.6. Changing the dataset does not significantly influence the overall detection results, and all four methods maintain consistent detection trends.

C. Simplified TCN Model Evaluation

Training Time: We continue to use Dataset 4 as the training data to evaluate the training process of our TCN model. Our comparison is with the original TCN model [16]. As shown in Fig.7(a), the simplified TCN has a slower convergence rate compared to the original TCN. The original TCN only needs about 80 epochs, and costs 40.63 seconds, to reach a stable loss. Under the same dataset and data segmentation method, our simplified TCN model requires 250 epochs and costs 74.14 seconds, to stabilize the loss to the same level as the original TCN. However, due to the simplified residual blocks, the training speed per epoch has increased, so the overall training speed of the simplified TCN model has not slowed down too much.

Detection Accuracy: As shown in Fig.7(b), the simplified TCN model did not suffer a decrease in detection accuracy.

Model Size: Our method of simplifying the TCN model structure significantly reduces the disk space requirement,

TABLE II: Time Latency of Hilbert-Algo

Cutoff	20%	23%	24%	25%
Latency	12s	13.8s	14.4s	15s
VIV Detection	Not detected	Not detected	True	True

TABLE III: System endurance with $0.5kW \cdot h$ battery

	VIV Detection	Endurance (hour)
KF	1.0	71.4286
AF (Ours)	1.0	101.2658

allowing the entire algorithm to be deployed on more resource-constrained edge devices. Under the same settings, the parameter file of the original TCN model occupies $63KB$, but ours only occupies $25KB$, which is a **60%** reduction in memory usage compared to the original TCN model. This enables more edge devices, such as STM32, to run our algorithm, potentially reducing costs in real-world sea-crossing bridge deployments.

D. TCN-based VPD Performance

Inference Latency: Compared to the current state-of-the-art white-box method, the Hilbert transform-based VIV detection method, the deep learning approach significantly benefits in terms of latency. The Hilbert transform-based VIV detection algorithm (Hilbert-Algo) has end effects [7], meaning the data at the beginning and end of the data window can render the Hilbert-Algo ineffective. To address this issue, [3], [7] cuts off the data to ensure detection effectiveness, typically causing at least 15 seconds latency, shown in Table.II using 0.2Hz VIV data with 1-minute window size in Dataset 3. Our TCN-based VPD method only requires a delay of no more than half the vibration period to detect VIV, which is at most about 2 seconds.

Computation Time Comparison: Finally, we compared the computation times of the TCN-based VPD method (Ours) and Hilbert-Algo. When processing the same 3 minutes (9000 data points) of VIV data by one CPU core, our method inference time is 0.016 seconds, while the Hilbert-Algo took 0.022 seconds. Our TCN-based VPD method is slightly faster.

E. Energy Efficiency

Shown in Table.III, the AF (Ours) offers an advantage in terms of power consumption. Compared to the KF, the overall system sustainability improves by **41.8%**.

Fig.8 shows the experiment result using Dataset 3. Both AF (Ours) and KF recognize all the VIV events. The GNSS is correctly activated in AF (Ours). In the initial 30 seconds of GNSS cold-start, the power consumption is slightly higher than in its continuous working state, with GNSS board maintaining around 2.1 watts. Above all, our adaptive method significantly reduces energy consumption.

V. LIMITATION AND FUTURE WORK

Although our hardware and software are designed for eventual deployment on target bridges, the current testbed and data

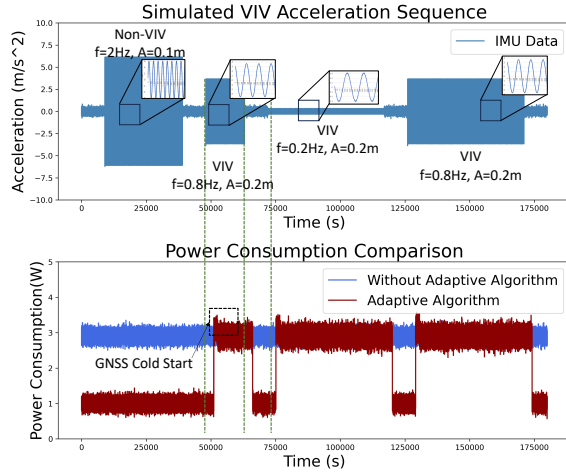


Fig. 8: Simulated IMU data with power consumption for GNSS and IMU System, testing the adaptive method efficiency and its positive impact on power consumption.

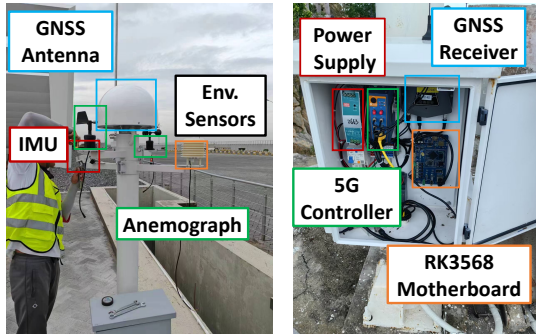


Fig. 9: Real deployment pictures of IoT-based bridge SHM station.

collection are not situated in an actual maritime environment. Experiments involving GNSS signal assessment and adaptive fusion were conducted separately under controlled conditions. While we have theoretically demonstrated the feasibility of serially integrating these processes, future work will focus on validating this approach on the target sea-crossing bridge. Additionally, we tested several classic models such as RNNs and CNNs, we ultimately selected TCN for our framework. However, newer models like time-series transformers [18], which require substantial data, were not explored in this study. This remains an area for future investigation.

In addition to addressing the aforementioned limitations, based on our testbed, an IoT-based bridge SHM system is in the process of deploying on a well-known sea-crossing bridge illustrated in Fig.9.

VI. CONCLUSION

This paper presents a **low-latency, high-accuracy, and energy-efficient method for bridge VIV detection**. The proposed method replaces the traditional VIV monitoring approach with TCN, significantly reducing detection latency. Additionally, we design an adaptive fusion algorithm for

GNSS-RTK and IMU data. By extracting and evaluating the multipath effects of GNSS in marine environments, our method maintains high VIV detection accuracy while substantially reducing the power consumption associated with GNSS devices, ensuring system endurance during emergency power outages. Finally, we develop an IoT-based bridge SHM system to deploy our algorithm in a real-world sea-crossing bridge.

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